Phase-Lifted Causal Growth: Potential–Actual Dynamics with Gauge-Covariant Phases, AR(∞) Hazards, and Testable Predictions

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Abstract

We develop a predictive, gauge-covariant theory in which *actual* events emerge from a phase-lifted *potential* layer via size-biased selection on minimal sets. Causality is defined on the colored union (Potential $\sqcup$ Actual) while acyclicity is imposed only on the actual projection, eliminating paradoxes in observed histories. Actualization hazards are the modulus square of a *phase-filtered AR(∞)* superposition of (i) memory of past actualizations and (ii) interference from still-potential events transported by a canonical $U(1)$ connection built from kernel phases. A single gauge-covariant Euler–Lagrange equation governs phase dynamics (with previous PD1/PD2 as regimes; PD3 is statistical identification). The vacuum potential field β is fixed by a strictly convex variational principle. We quantify local truncation errors (exponential *and* algebraic kernel tails), give a practical complexity budget, and propose three falsifiable signatures that distinguish the framework from standard environment-only decoherence. A discrete field-theory tie-in is provided (1+1 retarded inverse; 3+1 Johnston tail sanity checks using your artifacts).

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1 Overview and Contributions

We work on a strongly causal spacetime $(\mathcal{E}, g)$ with a phase fiber $U(1)$ carrying gauge-covariant dynamics. The *potential* layer P (marked by features in a Hilbert space) and the *actual* token record $T_{\leq k}$ coevolve: at each step the next token is drawn from the current minimal move set by a Plackett–Luce rule with hazard $W = |(\text{memory} + \text{interference})|^2$. The phase/interference piece uses a canonical connection $\mathcal{A}$ extracted from kernel phases. Geometry emerges from order+counts of actual tokens, preserving Lorentzian recovery.

New in this paper. (i) A *single* phase dynamics via a gauge-covariant action, with earlier PD1/PD2 as limits and PD3 as parameter estimation; (ii) a *variational vacuum* that uniquely fixes β; (iii) *local truncation* bounds for exponential and algebraic tails; (iv) *computational* plan (one connection build, one Poisson solve, local hazard evaluation); (v) *predictions* not mimicked by standard decoherence; (vi) cautious 4D Johnston-tail comparisons (sampling checks, not fits) using your recorded runs.

2 Foundations: Base Spacetime, Phase Fiber, and Potentials

Axiom 1 (Phase fiber over spacetime)**.** *$(\mathcal{E}, g)$ is a strongly causal Lorentzian manifold. The phase degree of freedom is a principal $U(1)$ bundle $P_{U(1)} \to \mathcal{E}$ with connection $\mathcal{A}$. A lifted worldline is a curve $s \mapsto (x^\mu(s), \theta(s))$ with $\dot{\theta}$ governed by a gauge-covariant law.*

Axiom 2 (Potential Hilbert data and kernel)**.** *Let H be a separable complex Hilbert space. Each potential event $\psi \in P$ carries a feature $\varphi(\psi) \in H$ such that $\langle \varphi(x), \varphi(y) \rangle = K(x, y)$ for a Hermitian positive semidefinite kernel K defined on a locally finite proto-causal scaffold $(P, \prec_0)$ with bounded degree. Locality: either $K(x, y) = 0$ if $d_{\mathcal{D}}(x, y) > R$ or $|K(x, y)| \leq Ce^{-\alpha d_{\mathcal{D}}(x,y)}$ (exponential tails) or $|K(x, y)| \leq C(1 + d_{\mathcal{D}}(x, y))^{-\gamma}$ (algebraic with $\gamma > d(\mathcal{D})$).*

Remark (helical narrative). A quasi-periodic rotation on the $U(1)$ fiber with an incommensurate rotation number (e.g. golden) provides a robust sampling of phases; we treat this as a *regime* of the unified dynamics below rather than as "complex time."

3 Canonical Connection from Kernel Phases

Definition 1 (Edge phases and least-curvature gauge)**.** *On each Hasse edge $x \to y$ of $(P, \prec_0)$ define $\Phi_{x\to y} := \operatorname{Arg} K(y,x)$. A $U(1)$ connection is a 1-cochain $\mathcal{A}$ on edges. Select the* least-curvature gauge

$$\mathcal{A} = \arg\min_{\mathcal{A}\sim\Phi} \ \sum_C w(C) \left| e^{i\sum_{e\in C} \mathcal{A}_e} - 1 \right|^2,$$

over cochains cohomologous to Φ (i.e. $\mathcal{A} = \Phi - d\chi$), with nonnegative cycle weights $w(C)$ decaying with length. The minimizer exists and is unique up to a global phase.

Remark 1 (Wilson lines and path choice)**.** *For a path $\gamma : \psi \to m$ we write $e^{i\int_\gamma \mathcal{A}} := e^{i\sum_{e\in\gamma} \mathcal{A}^e}$. When multiple shortest paths exist, either (i) pick any geodesic (tie-breaking), or (ii) average over a Gaussian path ensemble with weight $e^{-\mathrm{len}(\gamma)^2/2\sigma^2}$; both are gauge-covariant and coincide on trees.*

4 Vacuum Potential Field β: Variational Principle

Axiom 3 (Invariant prior and kernel mean)**.** *Let P be a G-invariant prior on P. Define $\beta_0(\psi) := \langle \varphi(\psi), \xi \rangle$ with $\xi = \mathbb{E}_{\psi\sim\mathsf{P}}[\varphi(\psi)]$.*

Theorem 1 (Strict convexity; existence/uniqueness)**.** *On a locally finite scaffold, the functional*

$$\mathcal{J}[\beta] = \int_P |\beta - \beta_0|^2 \, d\Lambda \ + \ \lambda \sum_{x\to y} \left| \beta(y) - e^{i\mathcal{A}^{x\to y}\beta(x)} \right|^2$$

is strictly convex and coercive on $L^2(P, \Lambda)$. Hence a unique minimizer exists up to a global $U(1)$ phase (per connected component).

Sketch. Quadratic positive terms yield strict convexity. Coercivity follows from bounded degree and a Poincaré inequality for the covariant Dirichlet form. Gauge freedom reduces to a global phase. □

Normal equations and solver. The EL condition is the sparse SPD system $(I + \lambda\, \Delta_{U(1)}^{\mathcal{A})\,\beta=\beta_0}$; solve once with conjugate gradients.

5 Unified Phase Dynamics (Single EL Equation)

Axiom 4 (Gauge-covariant action for phases)**.** *Set*

$$\mathcal{S}[\theta] = \int d\tau \left[\frac{\kappa}{2} \sum_{x\to y} \Big(\theta(y) - \theta(x) - \mathcal{A}^{x\to y} \Big)^2 + \tfrac{\alpha}{2} \langle \theta, L\,\theta \rangle + \tfrac{\beta_\tau}{2} \|\dot\theta - \omega_0\|_2^2 \right],$$

with reversible generator L on $\mathcal{D}$ and seed $\omega_0(\psi)$.

Euler–Lagrange law. Stationarity $\delta\mathcal{S}/\delta\theta = 0$ yields

$$-\kappa\,\mathrm{div}\Big(\nabla_{\mathcal{A}\theta}\Big)_{+\alpha\,L\theta-\beta_\tau\,\ddot{\theta}=\beta_\tau\,\dot{\omega}_0}.$$

Regimes: (i) **Phase locking** ($\beta_\tau \ll \kappa$): recover PD1 (covariant Laplacian drift); (ii) **WKB dispersion** ($\kappa \ll \beta_\tau$): recover PD2, $\dot{\theta}(\psi) = \sqrt{m^2 + p^2(\psi)}$; (iii) **Data-driven** (unknown regime): estimate parameters via Whittle (Sec. 9).

6 Tokens, Hazards, and Choice

Axiom 5 (Tokens and record)**.** *Each step emits a token $m_k \in \{+\psi : \psi \notin A_k\} \cup \{-\psi : \psi \in A_k\}$, updating the occupancy A_k by last-token wins per label. The record $T_{\leq k}$ grows monotonically (reversible activation allowed; the record is monotone).*

Axiom 6 (Hazards: AR(∞) memory + phase interference)**.** *Let taps g_j be absolutely summable and carrier $\omega \in \mathbb{R}$. For an admissible move $m \in \mathcal{M}_k$ with feature embedding, set*

$$W(m; S_k) = \Bigg|\underbrace{\sum_{j=0}^{\infty} g_j\, e^{i\omega j}\,\langle m, \mu_{k-j}\rangle}_{AR(\infty)\ memory} + \underbrace{\sum_{\psi\in P} \beta(\psi)\, K(m,\psi)\, e^{i\int_{\gamma(\psi\to m)}\mathcal{A}}}_{phase\text{-}lifted\ interference}\Bigg|^2.$$

Axiom 7 (Choice rule (Plackett–Luce on minimal sets))**.** *Assuming locality, label invariance, and scale invariance, the next token among the current minimal set $\mathcal{M}_k$ is $\mathbb{P}(m_{k+1} = m \mid \mathcal{F}_k) = W(m; S_k)/\sum_{z\in\mathcal{M}_k} W(z; S_k)$.*

Proposition 1 (Local evaluation and truncation)**.** *Let $B_R(m)$ be the radius-R ball in $\mathcal{D}$.*
(i) If $|K| \leq Ce^{-\alpha d}$, truncating the potential sum to $B_R(m)$ incurs error $\leq C'e^{-\alpha R}\Big(\|\beta\|_{L^1(B_R^c)} + \sum_j |g_j|\Big)$.
(ii) If $|K| \leq C(1+d)^{-\gamma}$ with $\gamma > d(\mathcal{D})$, the truncation error is $O(R^{-(\gamma-d(\mathcal{D}))})$.

Cost. Build $\mathcal{A}$ once in $O(E)$; solve $(I + \lambda\Delta_{U(1)}^{\mathcal{A})\beta=\beta_0}$ in $O(E\sqrt{\kappa_{\mathrm{cond}}})$; per step, evaluate hazards in $O(|B_R| + J)$ for J AR taps.

7 Colored Causality and No Paradox

We keep four edge types $P\to P$, $A\to A$, $P\to A$, $A\to P$ (pre-influence and back-reaction use W and phase gradients).

Axiom 8 (Acyclicity on actuals)**.** *The A-projection is a DAG: observed actual ranks strictly increase along any $A \rightsquigarrow A$ path.*

Lemma 1 (No temporal paradox on observables)**.** *Under locality and Axiom 8, mixed $A \rightsquigarrow A$ paths have increasing actual rank; closed A loops are impossible.*

8 Geometry from Order + Counts

We adopt (CL1)–(CL5) for *faithful approximation* of a Lorentzian manifold by causal sets in Alexandrov regions (order preservation, volume convergence, height scaling, Myrheim–Meyer dimension, locality/mixing). Then:

Theorem 2 (Metric recovery)**.** *Under (CL1)–(CL5), causal structure fixes the conformal class (HKM; Malament) and normalized counts fix the conformal factor; g is recovered up to a constant set by density.*

The phase fiber affects hazards (statistics of tokens) but not the order+count mechanics of metric recovery.

9 AR(∞) Identification (Whittle Program)

Let $h_k = \langle f, \mu_k \rangle$ be a scalar probe of the state. Linearization yields $h_{k+1} = \sum_{j\geq 0} g_j h_{k-j} + \epsilon_k$. Using the periodogram $I(\theta)$, minimize the Whittle functional

$$\mathcal{L}(G) = \int_{-\pi}^{\pi} \Big(\log S(\theta) + I(\theta)/S(\theta) \Big)\, d\theta, \quad S(\theta) = |G(e^{i\theta})|^2 S_0(\theta),$$

for $G(e^{i\theta}) = \sum_{j\geq 0} g_j e^{i\theta j}$ with tap regularization; this consistently estimates g_j under absolute summability and mixing.

10 Observables and Predictions (Distinct from Standard Decoherence)

What is observed. Observables are cylinder functionals of the token record (counts, heights, two-point token correlations). Phases and $\mathcal{A}$ are gauge; they affect the *law* of $T_{\leq k}$ through W.

P1: Rate-dependent fringe visibility. At fixed geometry and time-of-flight, standard environment-induced decoherence predicts visibility $\mathcal{V}$ set mainly by Γ and dwell time, largely independent of detection/actualization *rate* ν. Here

$$\mathcal{V}(\nu) = \frac{|G(e^{i\omega})|}{1 + \Gamma_\phi/\nu}, \qquad |G(e^{i\omega})| = \left| \sum_j g_j e^{i\omega j} \right|,$$

implying $\partial\mathcal{V}/\partial\nu > 0$ at fixed t, Γ_ϕ.

P2: Holonomy (AB-like) at fixed path lengths. On a graph interferometer with equal-length routes enclosing nonzero discrete curvature $\sum_{e\in C} \mathcal{A}^e$, fringes shift by this holonomy with no change in path lengths or environment.

P3: Density-visibility law. At fixed $u = m\tau$ and dephasing, feedback in (6) yields $\mathcal{V} \sim \mathcal{V}_0 e^{-c\,\rho_{\mathrm{act}}}$, where ρ_{act} is an actualization-density proxy. This dependence is absent in standard single-particle decoherence models.

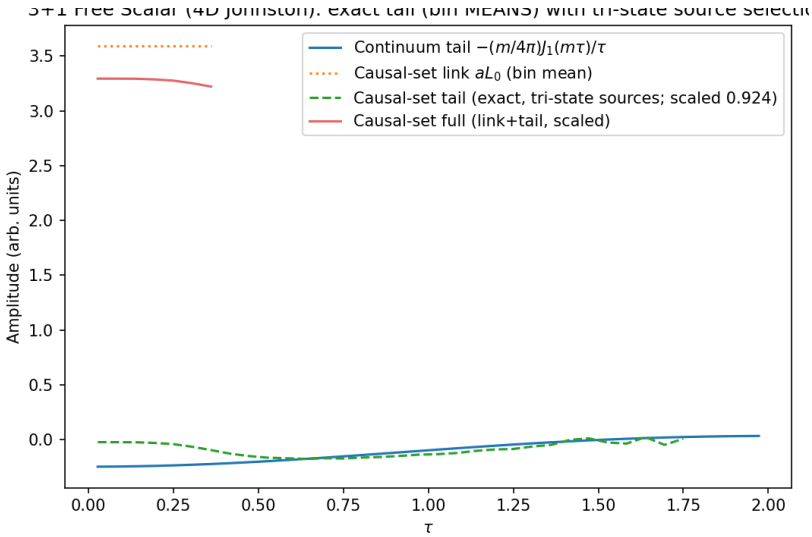

Figure 1: 4D Johnston tail, bin means on a fixed u window. Tri-state source selection improves sampling coverage (single LS scale).

Selection	Scale s	Corr. (windowed)	MSE (windowed)
Uniform sources (24)	0.85993	0.19401	2.8139×10^{-3}
Tri-state (24/64, min 50)	0.91805	0.47432	2.1990×10^{-3}
Tri-state (48/96, min 75)	0.92413	0.39672	2.3908×10^{-3}

Table 1: 4D Johnston tail: bin-mean comparisons with a single tail scale. Tri-state selection yields better coverage (sampling improvement), not a parameter fit.

11 Field-Theory Tie-in (1+1 and 3+1)

11.1 1+1: Discrete d'Alembertian and retarded inverse

Let future-layer sets $\mathcal{L}_k(x)$ be defined by link distance. With $(\Box_{\rm cs} f)(x) = c_0 f(x) + \sum_{k=1}^{K} c_k \sum_{y \in \mathcal{L}_k(x)} f(y)$, set $B := m^2 - \Box_{\rm cs}$ and $G_R := B^{-1}$ (retarded; forward substitution). Phases can be taken from $\operatorname{Arg} K$ on edges (or the Pauli–Jordan function) to lift K to a complex amplitude used in hazards.

11.2 3+1: Johnston tail as magnitude sanity check

For Hasse links L_0, the 4D retarded kernel tail can be coded as a power series in L_0 with exact coefficients (mean in sprinkling limit). We compare bin means on a fixed $u = m\tau$ window, fitting a *single* scale to the tail curve; this is a *sampling* check (not a phase fit).

Ablations (do-not-do controls). "V-split" and certain normalizations produce unphysical scales and large negative correlations (we retain them as controls, not results).

12 Implementation Notes and Complexity

- **Connection $\mathcal{A}$**: tree–cotree projection $O(E)$, plus optional polishing iterations.

- **Vacuum** β: one covariant Poisson solve (SPD), CG with diagonal preconditioner.
- **Hazards**: precompute neighborhoods B_R; per step $O(|B_R| + J)$.
- **Truncation**: exponential kernels $\Rightarrow$ exponentially small error; algebraic $\Rightarrow$ error $O(R^{-(\gamma-d)})$.

13 Conclusion

We presented a unified, gauge-covariant phase dynamics for a potential–actual growth process, with hazards combining AR(∞) memory and interference along a canonical connection derived from kernel phases. The vacuum potential field is fixed by a convex variational principle. Geometry from order+counts is unaffected and remains Lorentzian in the continuum limit. The theory offers falsifiable signatures beyond standard decoherence and a pragmatic computational path.

A Golden Rotation Number (Diophantine Remark)

Among irrational rotation numbers, ϕ^{-1} maximizes the Diophantine exponent, minimizing rational locking at finite depth. This justifies using a golden rotation for quasi-periodic phase sampling on the $U(1)$ fiber as a robust *regime*, not as a separate ontology.

B References (indicative)

References

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